

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and LangFlow”

Project Tile - Intelligent Shopping Agent

Domain of Project - E-commerce

Student Name	Recent Passport Photo	Email ID	Mobile number [WhatsApp]
Bintu Kadhiwala		bintu.kadhiwala@ scet.ac.in	9825564126

Problem Statement -

Problem Statement : Intelligent Shopping Agent

The Challenge

Shoppers today face an overload of product options, scattered reviews, and dynamic pricing across e-commerce platforms. Finding the right product that balances price, quality, and personal preferences is time-consuming and frustrating. Traditional recommendation systems often fail to capture real-time trends, personalized needs, or contextual factors like urgency, budget, or sustainability. The challenge is not just to recommend products but to act as a smart shopping companion that helps buyers make informed and confident decisions.

The Objective

Build an **agentic AI shopping assistant** that provides intelligent, personalized, and real-time support. The solution should:

- **Fuse Multisource Data** : Aggregate product details, reviews, ratings, and price comparisons across multiple e-commerce sites into unified insights.
- **Enable Multimodal Search** : Allow users to input voice queries, images (e.g., a photo of a product), or text descriptions to find best-matching products.
- **Create a Predictive & Agentic Layer** : Suggest purchase timing (e.g., discounts, seasonal offers), recommend alternatives, and alert users about price drops or fake reviews.
- **Visualize Smart Shopping Pulse** : Provide a dashboard summarizing comparisons, sustainability scores, trending products, and personalized shopping lists.

Proposed Solution -

The proposed solution **Intelligent Shopping Agent** stands out by combining **LangFlow**, **IBM watsonx.ai**, and **Granite Models** into a single intelligent system. It is developed to act as a smart shopping companion that helps buyers make informed and confident decisions. The system addresses challenges in finding the right product that balances price, quality, and personal preferences.

The system consists of following key agents:

- Multimodal Search Agent - Processes text, voice, and image inputs to identify and match products accurately.
- Shopping Companion Agent - Understands user needs and coordinates all shopping-related tasks and recommendations.
- Product Discovery Agent - Finds the most relevant products across multiple e-commerce platforms based on user preferences.
- RAG Knowledge Agent - Retrieves real-time product, review, and market information from trusted sources before generating responses.

The solution also includes a **user-friendly dashboard** (built with Streamlit or React) that visualizes comparisons, trends, price movements, and shopping summaries.

LangFlow is used to orchestrate agent workflows, while watsonx.ai ensures scalable model deployment, embeddings, and AI governance. Granite models power reasoning, summarization, and personalization.

Overall, this solution delivers an **end-to-end intelligent Intelligent Shopping Agent** that act as a smart shopping companion that helps buyers make informed and confident decisions besides recommending products.

Technology Used -

- **Langflow platform** - for designing and orchestrating multi-agent workflows visually
- **IBM Granite model** - for eg - ibm-Granite-4-h-small - for natural language understanding, reasoning, and personalization
- **IBM Cloud** - for providing the infrastructure to build, deploy, and scale the Intelligent Shopping Agent efficiently and enabling a **reliable, secure, and scalable environment** for deploying this Agent
- **RAG** - for fetching **real data first, then generate intelligent answers**, making the Intelligent Shopping Agent more trustworthy and effective.
- **IBM Watsonx.ai** - for model deployment, embeddings, and AI governance

Langflow component Used -

1. **Chat Input** - Interface where users enter their text queries, voice queries, images (e.g., a photo of a product), or text descriptions.
2. **IBM watsonx AI Agent** - Processes user input using granite models to recommend products and helps buyers make informed and confident decisions.
3. **Name of IBM Granite model** - Granite-4-h-small → Fast, efficient for real-time recommendations.
4. **Chat output** - Displays AI-generated responses based on real-time trends, personalized needs, or contextual factors like urgency, budget, or sustainability.

Langflow workflow -

← → ↺ ⓘ localhost:7860/flow/bc69a4b5-229a-4321-a610-7bd17ae9b357 🔑 ☆ 👤 School ⋮

 Starter Project /  Intelligent Shopping 150k 25k 🔔 

🔍 Search /

Components

🔗 Input & Output >

📄 Data Sources >

🛠 Models & Agents >

🧠 LLM Operations >

📁 Files & Knowledge >

⚙ Processing >

↔ Flow Control >

🔧 Utilities >

🔍 Discover more compone...

+ New Custom Component

🗨 Chat Input

🔗 IBM watsonx.ai

Generate text using IBM watsonx.ai foundation models.

Input

Type something... 🌐

System Message ⓘ

Type something... 🔄

watsonx API Endpoint* ⓘ

https://us-south.ml.cloud.ibm.c... 📄

watsonx Project_ID ⓘ

31ee1e08-f758-4ef9-8b0c-6ec71... 🌐

watsonx Space_ID ⓘ

Type something... 🌐

Watsonx API Key* ⓘ

..... 🌐 🔒

Model Name*

ibm/granite-4-h-small 📄

Language Model 📄

🛠 Agent

Define the agent's instructions, then enter a task to complete using tools.

Language Model* ⓘ

Receiving input 🔒

Agent Instructions ⓘ

You are a Langflow Agent — an AI 🔄

Tools ⓘ

Input ⓘ

Receiving input 🔒

Response 📄

🗨 Chat Output

🔗 77% ^ 📄 ? ⚙

🎮 Playground

Share ▾

🏠 🔍 Type here to search



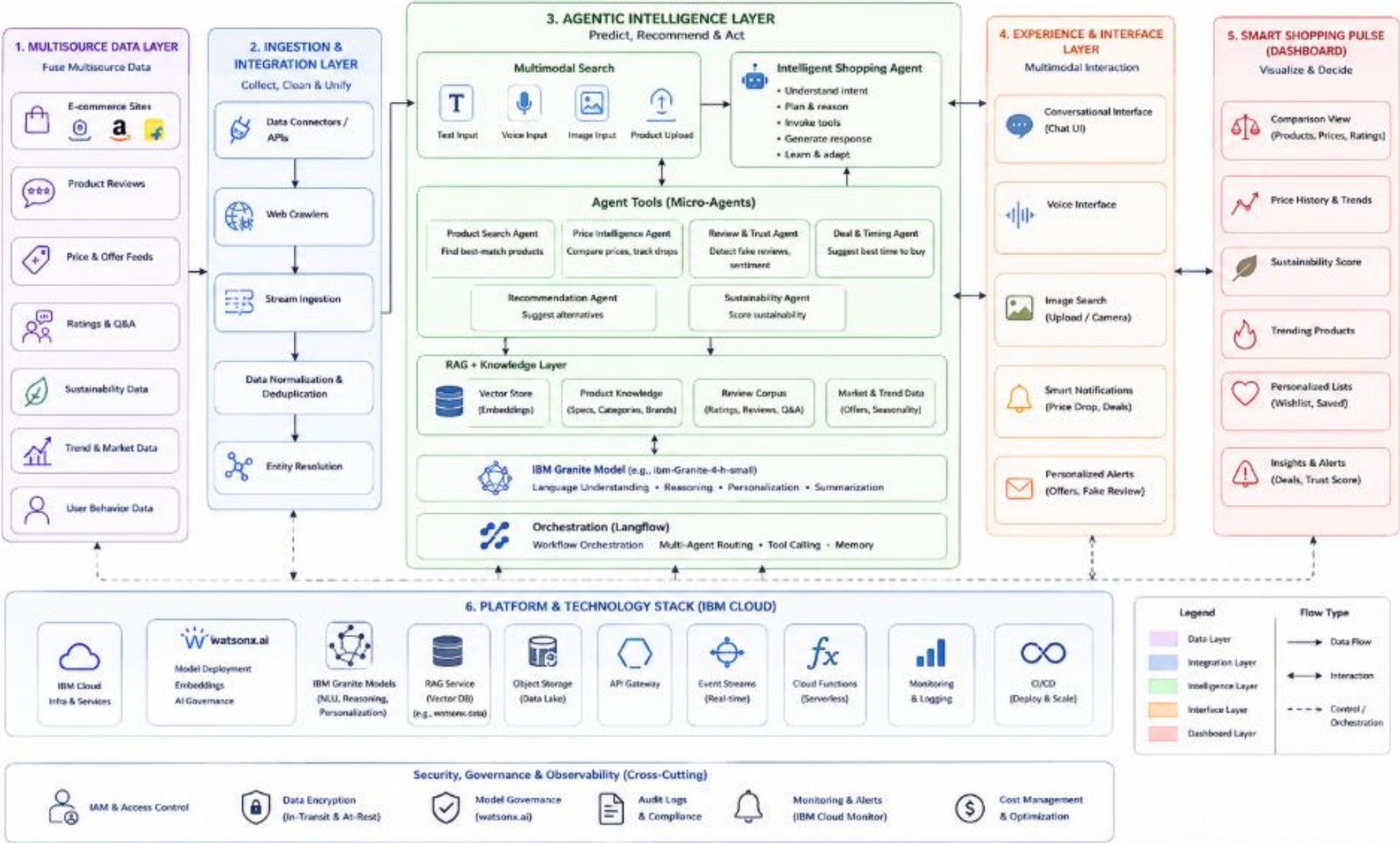
🌤 29°C Partly cloudy

⏮ 🖥 🔌 🔊 ENG 22:25 24-06-2026 6

Architecture Blueprint

Intelligent Shopping Agent – Architecture Blueprint

AI-Powered, Multimodal, Agentic Shopping Assistant

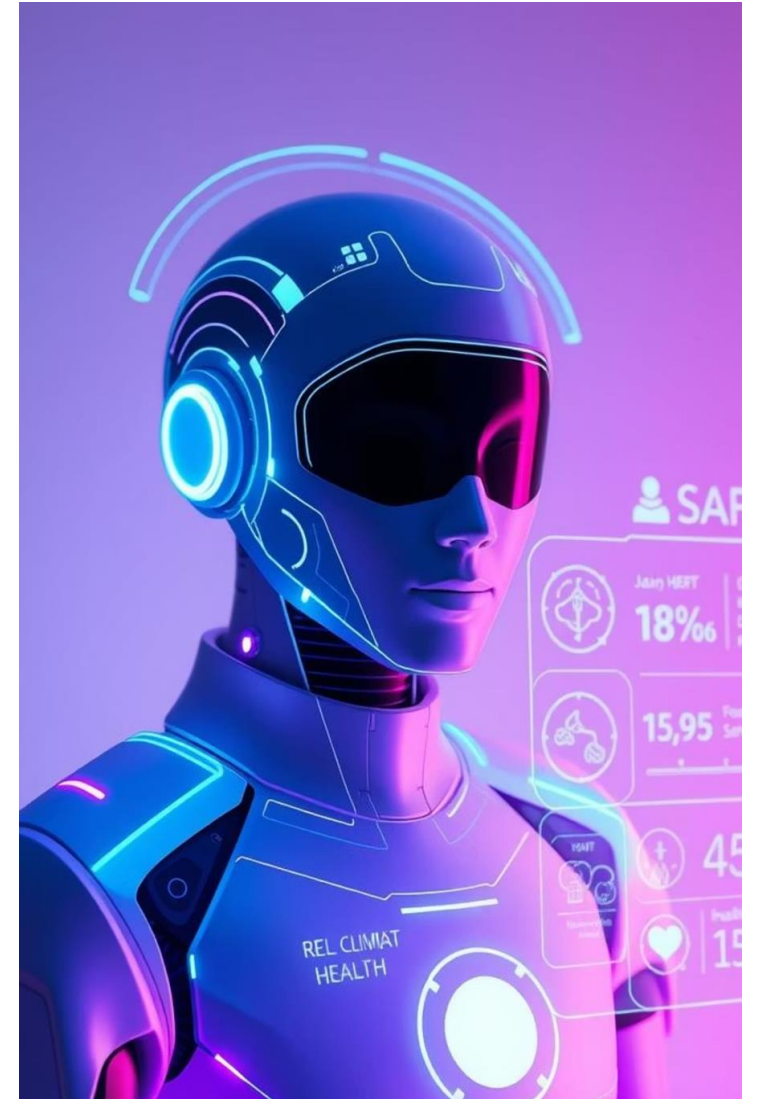


Role of Agentic AI in the solution

Role of Agentic AI in **Intelligent Shopping Agent**

- **Agentic AI** enables the Agent to function as an intelligent, autonomous system rather than a simple chatbot.
- Using **IBM watsonx.ai** and **IBM Granite models**, it coordinates multiple specialized agents to deliver personalized shopping guidance.
- Each agent performs a specific roles - such as retrieving required data, generating, etc.
- Agentic AI also brings **context awareness**, understanding user details like age, health conditions, and fitness goals to generate tailored recommendations.
- It supports **real-time adaptation**, continuously updating suggestions based on user inputs.
- Additionally, agents communicate with each other, enabling better decision-making and improved outcomes.

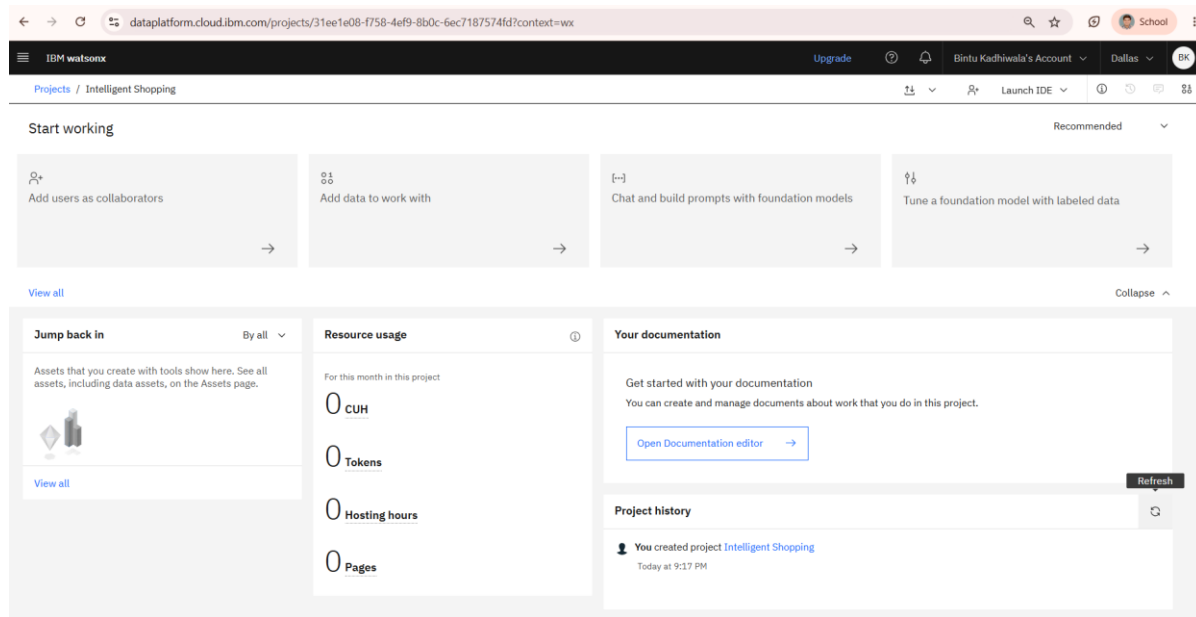
In gist, Agentic AI makes the system proactive, personalized, and goal-driven, acting like a **Intelligent Shopping Agent**.



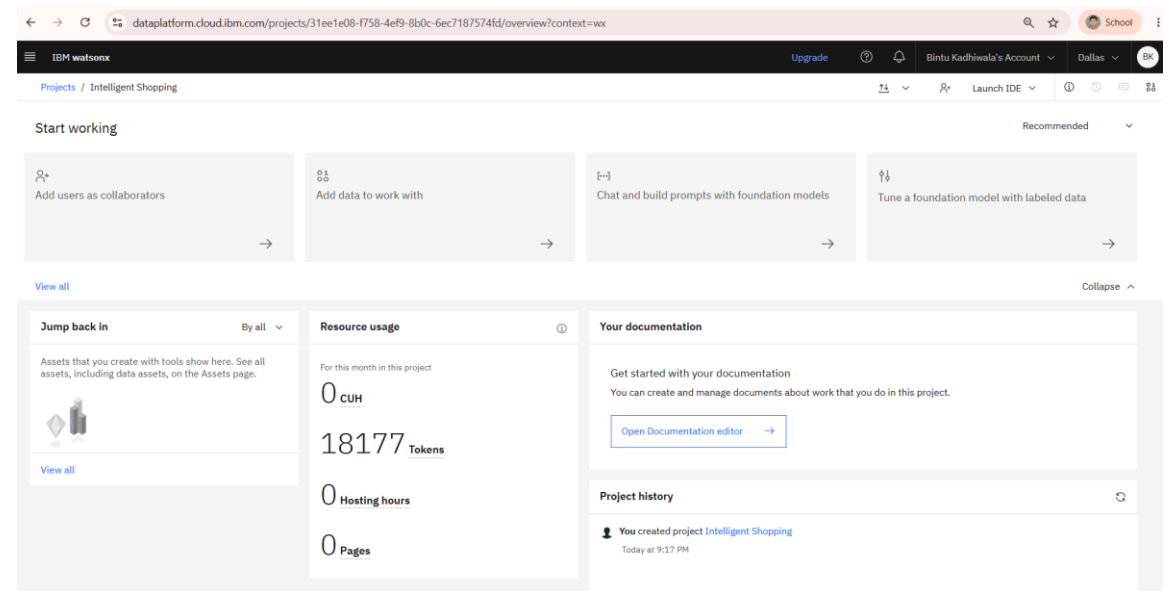
Token Usage

18,177 tokens are used

Before Prompt



After the Prompt



Novelty and Uniqueness :

The proposed **Intelligent Shopping Agent** stands out by combining **LangFlow**, **IBM watsonx.ai**, and **Granite Models** into a single intelligent system.

From Search Engine to Autonomous Shopping Agent - It actively assists decision-making throughout the purchasing journey besides responding to users' queries.

For example, "I am working as a professor at computer engineering department. can you suggest me which company laptop should i purchase dell laptop or hp laptop?"

Voice-based/Textual description-based discovery with Agentic Reasoning - It accepts voice and natural language descriptions and combines them with LLM reasoning to identify user intent.

For example, "I am working as a professor at computer engineering department. can you suggest me which company laptop should i purchase dell laptop or hp laptop?"

Create a Predictive and Agentic Layer - Suggest purchase timing (e.g., discounts, seasonal offers), recommend alternatives, and alert users about price drops or fake reviews.

For example, "suggest me the proper timings for purchase in the laptops of Dell company"

This makes the solution a **smart, adaptive, and proactive Intelligent Shopping Agent**.

Git Hub Link :

Working Github URL -

<https://github.com/bintukadhiwala/Intelligent-Shopping-Agent>

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Future Scope :

1. **Integration of Trust and Authenticity Analyzer** : Help users identify reliable products and sellers by generating a "Trust Score" based on reviews, seller reputation, return rates, and warranty coverage, highlighting verified customer experiences, analyzing seller credibility and fulfillment history, generating flag potentially counterfeit products, etc. **This feature reduces purchase risk and increases buyer confidence.**
2. **Budget Optimization and Cart Intelligence** : Help users maximize value within a budget by building optimal product bundles, suggesting cheaper alternatives with similar quality, etc. For example, a user with a ₹20,000 budget for a gaming setup receives an optimized combination of monitor, keyboard, mouse, and headset. **This feature increases customer satisfaction and purchasing efficiency.**
3. **Post-Purchase Assistant** : Continue assisting users after the purchase by tracking delivery status, monitoring return and warranty deadlines, recommending compatible accessories after purchase, suggesting setup guides and tutorials (if any), notifying users about recalls or product issues, etc. **This feature extends engagement beyond checkout and creates a complete shopping lifecycle assistant.**

Certificates Earned :

credly certificate :



Certificates Earned :

IBM BOB Certificate

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Certificates Earned :

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IBM **SkillsBuild**

Completion Certificate



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(ALM-COURSE_3825253)

According to the Adobe Learning Manager system of record

Completion date: 25 Jun 2026 (GMT)

Learning hours: 50 mins

Thank You!

Thank you for your time and interest.